

Friedrich Doku

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Research Focus

Quantum error correction and real-time decoding; FPGA/RTL implementation of surface-code decoders; decoder timing and power side channels; low-level systems and hardware security.

Education

Northwestern University

PhD Candidate Computer Science, Advisor: Peter Dinda

Evanston, IL

2024–Present

University of Pittsburgh

BS in Computer Science

Pittsburgh, PA

2023

Research Experience

Quantum Error Correction & Decoder Hardware, Northwestern University 2025–Present *Evanston, IL (with Prof. Kaitlin N. Smith)*

- Discovered a new side channel in fault-tolerant quantum computing: showed for the first time, on real hardware, that a classical QEC decoder’s execution time and power leak the error syndrome it processes.
- Ported the distributed union-find (Helios) and a constant-latency lookup-table decoder to a Xilinx Artix-7 FPGA and built the full measurement pipeline, capturing per-decode timing and power across hundreds of thousands of decodes of real Google Willow surface-code syndromes (distances 3 and 5).
- Fingerprinted individual physical quantum devices (IBM Herons) from decoder timing alone with up to 89% accuracy, and reconstructed code distance, logical error rate, and detector-firing structure from timing and power distributions.
- First author of “[Hardware Fingerprinting FTQC via Quantum Decoder Timing](#)”; follow-on work (DAC submission) in progress.

CAPIO: Safe Kernel-Bypass with Capabilities, Northwestern University

2024–Present

- Developed a capability-based kernel-bypass design for commodity NIC and NVMe devices, enabling direct userspace access to selected MMIO registers while isolating privileged registers on the same page.
- Built low-level kernel and device-driver mechanisms and mentored students working on systems implementation.

TrustCheckpoints: Keyless Software Attestation, Northwestern University

2024–Present

- Developed a software-only root-of-trust design that requires no on-device secret keys or specialized trusted hardware.
- Implemented randomized polynomial memory challenges and timing-based malware detection on ARM Cortex-A53 hardware.
- Built an end-to-end prototype spanning firmware, hypervisor, Linux, and an external verifier.

Privacy Backplane, University of Pittsburgh / Northwestern / University of Michigan

2022–2025

- Developed systems mechanisms for IoT data-privacy control, including an ARM/Linux video pipeline and kernel-level components.

Publications & Preprints

- **F. Doku**, J. Szefer, K. N. Smith. “Hardware Fingerprinting FTQC via Quantum Decoder Timing.” [arXiv:2609.12145](#), 2026.
- **F. Doku** et al. “CAPIO: Safe Kernel-Bypass of Commodity Devices using Capabilities.” *Workshop on Runtime and Operating Systems for Supercomputers (ROSS), SC26: The International Conference for High Performance Computing, Networking, Storage, and Analysis*, 2026. [arXiv:2512.16957](#). **To Appear.**
- **F. Doku** et al. “TrustCheckpoints: Time Betrays Malware for Unconditional Software Root of Trust.” [arXiv:2509.22762](#), 2025.
- **F. Doku** et al. “Khopesh: Contact Tracing Without Sacrificing Privacy.” *International Conference on Security and Privacy in Communication Systems (SecureComm)*, 2020. [Springer](#).

Technical Skills

Quantum / Decoding: surface codes, union-find, MWPM, lookup-table decoders, Qiskit, Stim

FPGA / Hardware: Verilog, Xilinx Artix-7, Vivado, ChipWhisperer, embedded ARM, PCBs

Systems: Linux kernel, hypervisors, device drivers, GDB, Buildroot, Yocto, U-Boot, BusyBox

Languages: Python, C, C++, Go, Java, Verilog, ARM/x86 assembly

Tools: Docker, Kubernetes, AWS, Oracle Cloud, Bash, PyTorch, TensorFlow

Industry Experience

Reverse Engineer, Red Balloon Security Dec 2023–Jul 2024

- Reverse-engineered ARM firmware and embedded systems to identify security vulnerabilities and developed RASPUTIN, an automated reverse-engineering system.

TITANS Cybersecurity R&D Intern, Sandia National Laboratories May 2022–Apr 2023

- Developed kernel- and hypervisor-level software for ARM systems in a cybersecurity R&D environment.

Software Engineering Intern, Carnegie Robotics May 2023–Jul 2023

- Developed embedded-platform test software for camera sensors and wrote kernel-level ARM code; worked with PCBs and hardware automation.

Software Engineer Contractor, Whonix Aug 2021–Sep 2022

- Built kernel-module-signing and live-boot functionality for a privacy-focused Linux distribution.

Software Engineer Intern, Oracle Corporation May 2021–Aug 2021

- Developed a secure IoT data store using Oracle Cloud and trusted hardware; worked with embedded Linux, BusyBox, U-Boot, Docker, C, and Java.

Earlier Research

University of Pittsburgh — secure embedded image processing, device drivers, IPC, and Hafnium hypervisor development. 2020–2023

Awards

Chancellor's Undergraduate Research Fellowship, University of Pittsburgh 2020